SESSION 6

FINANCIAL PLANNING & ANALYSIS – 2

SCALER SCHOOL OF TECHNOLOGY

Learning Outcomes

Session 6: Cost-Volume-Profit (CVP) Analysis

By the end of the session, learners will be able to:

- **Explain** the concept and importance of Cost-Volume-Profit (CVP) analysis.
- **Identify** the relationship between costs, volume, sales and profit.
- **Calculate** contribution, P/V ratio, break-even point and margin of safety.
- **Analyse** the impact of changes in price, costs and volume on profitability.
- **Evaluate and apply** CVP analysis to support pricing, production and cost-control decisions.

Cost-Volume-Profit (CVP) Analysis

Full /Traditional Costing: Also called **absorption costing**. Under this method, **both fixed and variable production overheads** are **included** in the **product cost.**

✔ **Fixed production overheads** are **absorbed** into the **unit cost.**

Sales	XXX
Less: Cost of goods sold *(Cost of production + Opening stock − Closing stock)*	(XXX)
Gross profit *(Sales − Cost of goods sold)*	**xxx**
Add/(Less): Over/(under)-absorbed OH *Actual fixed production OH vs. amount absorbed*	XXX
Less: Selling & admin overhead *Fixed + variable*	(XXX)
Net profit	**xxx**

Cost-Volume-Profit (CVP) Analysis

Marginal Costing: Also called **variable costing,** prepares the profit statement by **separating variable costs** from **fixed costs,** rather than by function (production, admin, selling) as in absorption costing.

✔ **Fixed costs** are treated as **period costs** and charged in full to the period, not spread over units produced.

Sales	XXX
Less: Variable cost	(XXX)
Contribution *Sales – Variable cost*	**XXX**
Less: Fixed cost	(XXX)
Profit *Contribution – Fixed cost*	**XXX**

Cost-Volume-Profit (CVP) Analysis

Cost–Volume–Profit (CVP) Analysis is a management accounting technique used to understand **how changes in costs, sales volume, selling price, and profit** affect a business.

It rests on **separating costs into fixed and variable components,** then using the resulting **cost-volume-profit relationships** to find the **break-even point** and beyond.

It helps managers answer questions such as:

- How many units must we sell to **break even?**
- How many units must we sell to earn **a target profit?**
- What happens to profit if **sales volume increases or decreases?**
- What happens to profit if the **selling price changes?**
- What happens to profit if **variable or fixed costs change?**
- How much sales can fall before the business starts making a loss?

Cost-Volume-Profit (CVP) Analysis

Key Terms: is a management accounting technique used to understand **how changes in costs, sales volume, selling price, and profit** affect a business.

It rests on **separating costs into fixed and variable components,** then using the resulting **cost-volume-profit relationships** to find the **break-even point** and beyond.

- **Selling price (SP)** — revenue per unit
- **Variable cost (VC)** — cost per unit that changes with volume
- **Contribution margin (CM) = Selling Price − Variable Cost** — the amount each unit contributes toward covering fixed costs and then profit
- **Fixed costs (FC)** — costs that don't change with volume (rent, salaries, etc.)
- **Break-even point (BEP)** — the volume at which total revenue equals total costs (zero profit)

Break-even analysis is a management accounting technique used to determine the **sales level at which total revenue equals total costs.**

At the **break-even point (BEP):**

Total Revenue = Total Cost (SP)

Therefore, **Profit = 0 (the business makes neither a profit nor a loss)**

Cost-Volume-Profit (CVP) Analysis

Key formulae used in CVP & Break-even Analysis

1. Sales Revenue: Total revenue generated from selling the products.

$$\text{Sales Revenue} = \text{Selling Price per Unit} \times \text{Units Sold}$$

Example: If the selling price per unit is ₹2,000 and the total units sold are 5,000 units, then
Sales Revenue = ₹2,000 x 5,000 = ₹1,00,00,000

2. Total Variable Cost: Total cost of variable inputs that changes with the level of activity.

$$\text{Total Variable Cost (TVC)} = \text{Variable Cost per Unit (V)} \times \text{Number of Units (Q)}$$

Example: A manufacturer produces 10,000 units.

- Variable material cost/unit = ₹1,200
- Variable labour cost/unit = ₹500
- Variable overhead/unit = ₹300

Variable Cost/Unit = ₹1,200+₹500+₹300=₹2,000
Total Variable Cost (TVC) = ₹2,000 x 10,000 = ₹ 2,00,00,000

Cost-Volume-Profit (CVP) Analysis

Key formulae used in CVP & Break-even Analysis

3. Contribution: It is the amount remaining after variable costs have been deducted from sales.

$$\text{Contribution} = \text{Sales} - \text{Variable Cost}$$

Therefore,

$$\text{Total Contribution} = (\text{Selling Price per Unit} - \text{Variable Cost per Unit}) \times \text{Units Sold}$$

Or,

$$\text{Total Contribution} = \text{Contribution per Unit} \times \text{Units Sold}$$

Example: A company sells 10,000 units of a product.

- Selling price per unit = ₹500
- Variable cost per unit = ₹300

Contribution Per Unit = ₹500 – ₹300 = ₹200
Total Contribution = ₹200 x 10,000 = ₹ 20,00,000

Cost-Volume-Profit (CVP) Analysis

Key formulae used in CVP & Break-even Analysis

4. Profit: It is the difference between total contribution and fixed costs.

$$\text{Profit} = \text{Total Contribution} - \text{Total Fixed Cost}$$

5. Profit – Volume (P/V) Ratio: The P/V ratio measures **contribution as a percentage of sales.**

✔ The **P/V ratio** indicates the **contribution** generated from **every ₹1 of sales.**

$$\text{P/V Ratio} = \frac{\text{Contribution}}{\text{Sales}} \times 100$$

Example: A company sells a product with the following details.

- Selling price = ₹500
- Variable cost = ₹300
- Contribution = ₹500 − ₹300 = ₹200

Therefore, P/V Ratio = $\frac{₹200}{₹500} \times 100$

= 40%

It means that for **every ₹100 of sales, the company generates ₹40 of contribution.**

Cost-Volume-Profit (CVP) Analysis

Key formulae used in CVP & Break-even Analysis

6. Break-Even Point (BEP) – in Units: The **break-even point (BEP) in units** is the exact **number of products** a business must **sell to cover all its costs and make zero profit or loss.**

$$\text{BEP Units} = \frac{\text{Fixed Cost}}{\text{Contribution per Unit}}$$

6. Break-Even Point (BEP) – in Sales Value: Sometimes management wants to know **the rupee value of sales required to break even,** rather than the number of units.

$$\text{BEP Sales} = \frac{\text{Fixed Cost}}{\text{P/V Ratio}}$$

Example: A company has:

- Fixed Cost = ₹2,00,000
- P/V Ratio = 40%

Therefore, **BEP (Sales Value)** = $\frac{₹2,00,000}{40\%}$

BEP (Sales Value) = ₹5,00, 000

Cost-Volume-Profit (CVP) Analysis

Key formulae used in CVP & Break-even Analysis

7. Margin of Safety (MoS) – in Units: A company's **margin of safety** is the **difference between its current sales and its break-even sales.**

- ✔ The margin of safety tells the company **how much it could lose in sales before the company begins to lose money.**
- ✔ In other words, before the company falls below the break-even point.
- ✔ The higher the margin of safety is, the lower the risk is of not breaking even or incurring a loss.

$$\text{MoS (Units)} = \text{Actual Units Sold} - \text{Break-Even Units}$$

Example: A company sells 10,000 units and has a break-even point (in units) of 6,000.

Therefore, MoS = 10,000 – 6,000 = 4,000 units.

So, the **4,000-unit margin of safety** represents the **company's cushion against a fall in sales volume.**

Cost-Volume-Profit (CVP) Analysis

Key formulae used in CVP & Break-even Analysis

8. Margin of Safety (MoS) – in Sales Value:

$$\text{MoS (Value)} = \text{Actual Sales} - \text{Break-Even Sales}$$

It can also be **expressed as a percentage:**

$$\text{MoS \%} = \frac{\text{MoS}}{\text{Actual Sales}} \times 100$$

Example: A company's sales are as follows:

- Actual Sales = ₹10,00,000
- Break-Even Sales = ₹6,00,000

Therefore, MoS = ₹10,00,000 – ₹6,00,000 = ₹4,00,000

$$\text{MoS \%} = \frac{₹4,00,000}{₹10,00,000} \times 100$$

$$\text{MoS \%} = 40\%$$

This means actual **sales can decline by 40%** before the business **reaches BEP**.

Cost-Volume-Profit (CVP) Analysis

Key formulae used in CVP & Break-even Analysis

9. Sales Volume Required to Achieve a Target Profit

$$\text{Required Sales Volume (Units)} = \frac{\text{Fixed Costs} + \text{Target Profit}}{\text{Contribution per Unit}}$$

Example: A company's reported sales and cost figures are as follows:

- Selling price per unit = ₹500
- Variable cost per unit = ₹300
- Fixed costs = ₹12,00,000
- Target profit = ₹8,00,000

The company wants to know **how many units it must sell to earn ₹8,00,000 profit.**

Contribution Per Unit: = ₹500 − ₹300 = ₹200

Required Sales to achieve target profit of ₹8,00,000:

$$\frac{₹12,00,000 + ₹8,00,000}{₹200}$$

= 10,000 units

Cost-Volume-Profit (CVP) Analysis

Illustration:

Say a company sells a product at:

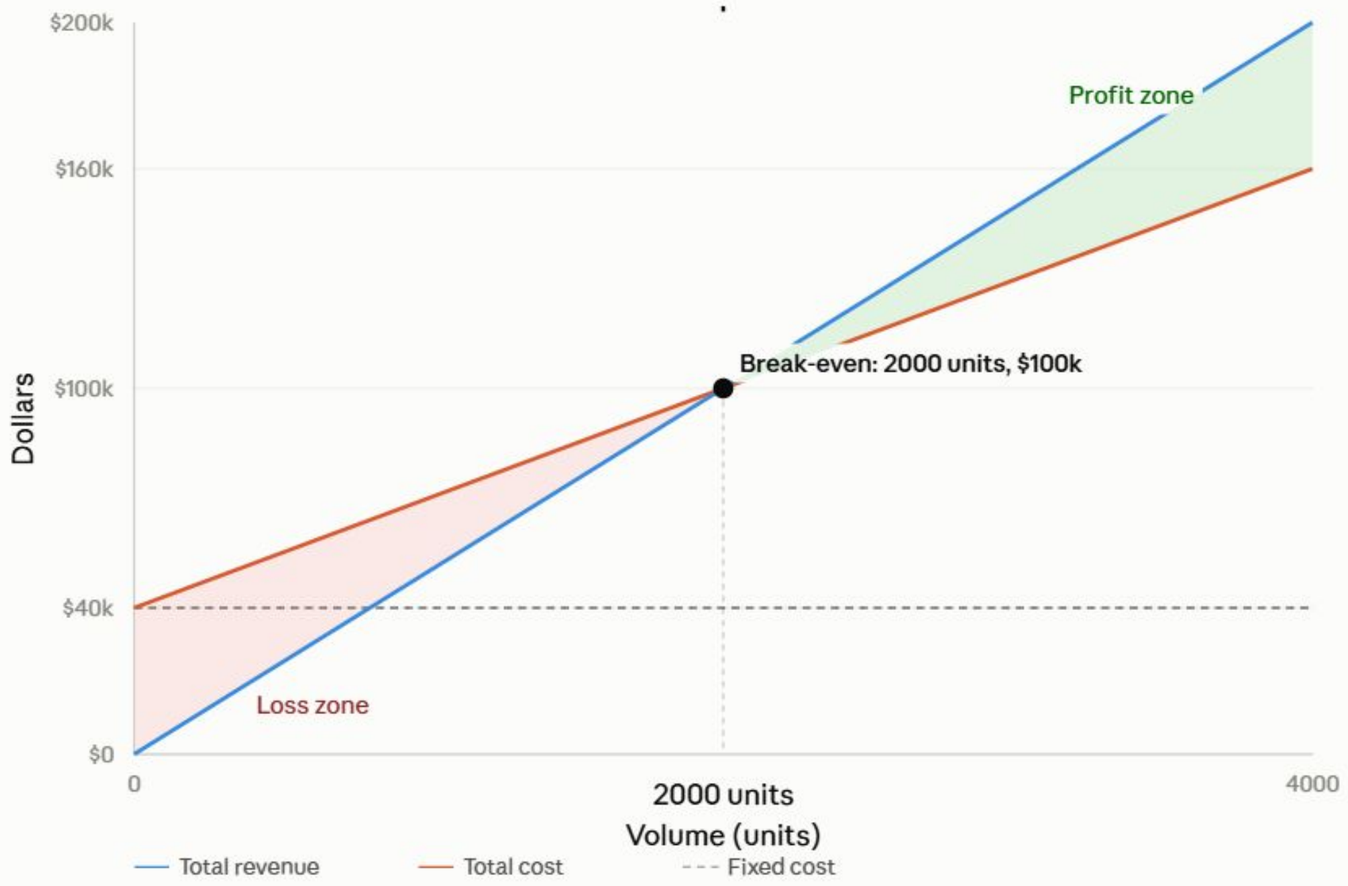
- Selling price = $50/unit
- Variable cost = $30/unit
- Fixed costs = $40,000

Contribution margin = $50 − $30 = $20/unit

BEP (units) = $40,000 ÷ $20 = **2,000 units**
BEP (sales value) = 2,000 × $50 = **$100,000**

Illustration:

Below 2,000 units the company operates at a loss (total cost exceeds revenue); above 2,000 units, every additional unit adds $20 of profit.



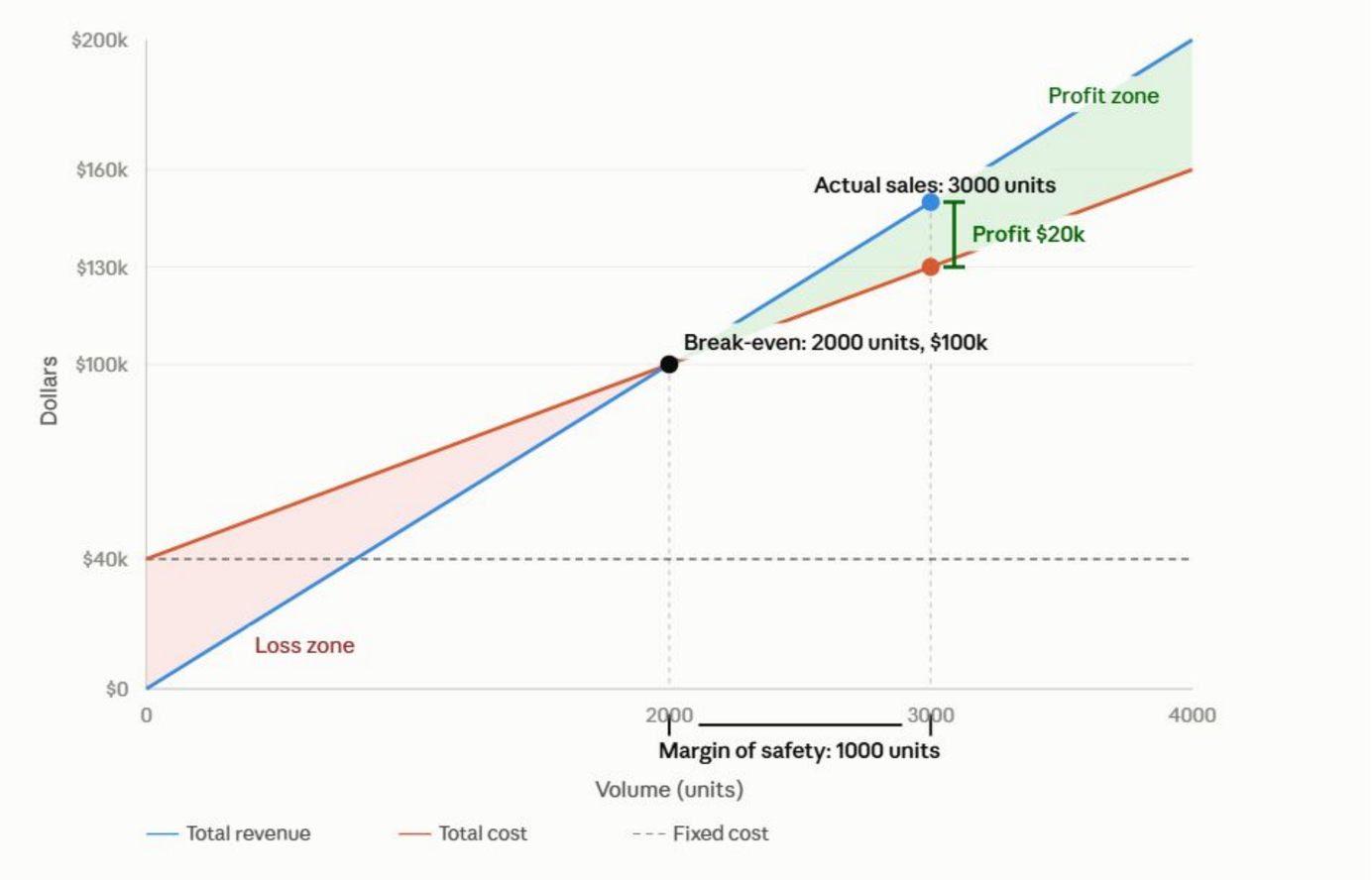
- The **revenue line (blue) starts at $0 and rises steeply** — every unit sold adds $50.
- The **total cost line (orange) starts at $40,000** (the fixed cost) and rises more gently — every unit adds only $30.
- The **fixed cost line (dashed gray) stays flat** at $40,000 regardless of volume.
- Where the revenue and total cost lines cross is the break-even point: 2,000 units, $100,000 in sales.
- Below that volume, **the loss zone (red) shows total cost exceeding revenue.**
- Above it, the **profit zone (green) shows revenue pulling ahead** — and the gap between the two lines widens by $20 for every extra unit sold.

Illustration:

Extending the example — suppose the company actually sells 3,000 units (instead of just breaking even at 2,000):

- Revenue = 3,000 × $50 = $150,000
- Total cost = $40,000 + (3,000 × $30) = $130,000
- **Profit = $150,000 − $130,000 = $20,000**
- Margin of safety = 3,000 − 2,000 = **1,000 units**, or $50,000 in sales
- MOS ratio = $50,000 ÷ $150,000 = **33.3% —** sales could drop by a third before the company starts losing money

Illustration:



- The **red triangle is the loss zone** — its height at any volume below 2,000 units is exactly how much the company is losing at that level of sales.
- The **green triangle is the profit zone —** its height at any volume above 2,000 units is the profit at that level.
- At 3,000 units (the actual sales point marked), the gap between the blue revenue line and the orange cost line is $20,000 — **that's the profit,**
- The **black bracket on the x-axis between** the break-even point and the actual sales point is **the margin of safety: 1,000 units, or $50,000 of sales cushion** before the company would start losing money again.

Thank You!